

QIRAN JIA

MACHINE LEARNING SCIENTIST · STATISTICIAN

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SUMMARY

ML research scientist (Biostatistics PhD candidate) with **5+ years** building statistical, variational Bayes, and deep-learning methods that turn **high-dimensional, multimodal data** with **complex correlation structure** into reliable scientific discoveries, actionable causal insights, and pragmatic predictions. Led projects from problem framing through modeling, inference, and large-scale benchmarking to adopted software, with **11** peer-reviewed papers (**7** methodological, **4** applied). Mentor junior PhD students and, as a statistical consultant, work with cross-functional collaborators to tailor data analysis workflows and translate technical approaches and findings for broad audiences

EDUCATION

University of Southern California <i>Ph.D. in Biostatistics</i>	Los Angeles, CA Jan. 2027 (expected)
New York University <i>M.S. in Biostatistics</i>	New York, NY May 2022
University of California, Santa Barbara <i>B.A. in Statistics and Chemistry</i>	Santa Barbara, CA Dec. 2019

INDUSTRY EXPERIENCE

Cellanome <i>COMPUTATIONAL BIOLOGY INTERN</i> <i>Cellanome's R3200 platform produces data that pairs longitudinal live-cell imaging with endpoint transcriptomics in each cell enclosure</i>	San Diego, CA Jun. 2026 - Aug. 2026
• Trajectory representation learning: Built a deep-learning trajectory autoencoder compressing imaging sequences into dynamic cell embeddings (10,000+ cells per run); achieved a 60× speedup over statistical baselines by leveraging GPU; adopted in internal software • Integrative multimodal modeling: Fused dynamical imaging embeddings with endpoint transcriptomics, improving cell-fate prediction and treatment association over baselines and revealing novel morphodynamical cell states missed by a snapshot modality • Upstream computer vision: Partnered with AI team on the platform's image-preprocessing and cell-segmentation workflows to maximize downstream data quality; built rigorous quality-control and integrative analysis pipelines for data from multiple novel experiments • Engineering practices: Leveraged agentic AI coding tools to speed delivery; reviewed teammates' code routinely	

RESEARCH EXPERIENCE

University of Southern California <i>GRADUATE RESEARCH ASSISTANT, BIOSTATISTICS</i>	Los Angeles, CA Aug. 2022 - Present
• Bayesian transfer learning: Developed Praxis-BGM, a variational Bayesian clustering method implemented in JAX (<i>Bioinformatics</i> 2026) borrowing structure from large references for small cohorts; in two real data applications, beat XGBoost (71.4% vs. 63.8%) and reference mapping methods (97.5% vs. 91.5%) • Integrative clustering: Co-developed LUCID, a quasi-mediation and prediction method and R software (<i>R Journal</i> 2024) for multi-omics with exposures and outcomes; extended missing-data handling, flexible integration strategies, and a 120× speedup (36,000+ downloads) • High-dimensional causal mediation: Developing a scalable variational Bayes extension of LUCID in JAX that formalizes latent-cluster causal mediation for high-dimensional exposures and multi-omics; outperformed benchmarks in simulated signal detection • Applied analyses: Performed integrative data analyses linking environmental exposures to high-dimensional molecular mediators and health outcomes in 4 applied papers (<i>Environment International</i>, <i>Communications Medicine</i>, <i>Environmental Pollution</i>, <i>Sci. Total Environ.</i>) • Leadership & mentorship: Mentor 2 first-year PhD students in weekly 1:1s on data analysis and statistical & ML method development; statistical consultant for clinicians and epidemiologists, translating research questions into rigorous analyses	
New York University <i>GRADUATE RESEARCH ASSISTANT, BIOSTATISTICS</i>	New York, NY Feb. 2021 - Sep. 2022
• Computer vision: Proposed BiTr-UNet, a CNN-Vision Transformer for 3D multimodal MRI brain tumor segmentation that models local and global spatial context; outperformed state-of-the-art methods then on the BraTS 2021 dataset (median Dice 92.6%, 93.5%, 88.7%) • Multiple testing under dependence: Co-developed DeepFDR (AISTATS 2024), a CNN-based framework that models spatial dependence to increase power while controlling FDR in 3D neuroimaging; detected 1,087 significant voxels distinguishing early-MCI converters from controls, versus none identified by five benchmark methods • Adversarial attack & defense: Co-authored MFI-PSO (<i>IJCNN</i> 2022), crafting targeted adversarial images via influence-guided particle swarm optimization; retraining models on them strengthened defense, cutting vulnerable test images by 56–59% (best of 6 methods)	

SKILLS

Programming Python, R, SQL; JAX, PyTorch, NumPy, Scanpy, Seurat, ggplot2; Linux, Git
Statistical Methods Bayesian inference, variational inference, dimension reduction, causal inference, integrative clustering, longitudinal data analysis, survival analysis, missing data, large-scale multiple testing under dependence, experimental design
Machine Learning Representation learning, self-supervised learning, computer vision, transformer-based models, transfer learning
Computational Biology Multi-omics integration, transcriptomics, cell imaging and morphology, statistical genetics
Computing GPU/CUDA, Unix & cluster-computing environment, scientific software development

SELECTED PUBLICATIONS

Praxis-BGM: Clustering of Omics Data Using Semi-Supervised Transfer Learning for Gaussian Mixture Models via Natural-Gradient Variational Inference.

Jia Q, Goodrich JA, et al.
Bioinformatics, 2026.

DeepFDR: A Deep Learning-Based False Discovery Rate Control Method for Neuroimaging Data.

Kim T, Shu H, Jia Q, de Leon MJ, et al.
Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR, 2024.

A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data.

Kim T, Jia Q, de Leon MJ, Shu H; Alzheimer's Disease Neuroimaging Initiative.
Medical Image Analysis, 2026.

MFI-PSO: A Flexible and Effective Method in Adversarial Image Generation for Deep Neural Networks.

Shu H, Shi R, Jia Q, Zhu H, Chen Z.
International Joint Conference on Neural Networks (IJCNN), 2022.

BiTr-UNet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation.

Jia Q, Shu H.
MICCAI BrainLes Workshop, 2021.

LUCIDus: An R Package for Implementing Latent Unknown Clustering by Integrating Multi-omics Data With Phenotypic Traits.

Zhao Y, Jia Q, Goodrich JA, Conti DV.
The R Journal, 2024.

An Extension of Latent Unknown Clustering Integrating Multi-omics Data (LUCID) Incorporating Incomplete Omics Data.

Zhao Y, Jia Q, Goodrich JA, Darst B, Conti DV.
Bioinformatics Advances, 2024.

Integrating Multi-Omics with Environmental Data for Precision Health: A Novel Analytic Framework and Case Study on Prenatal Mercury-Induced Childhood Fatty Liver Disease.

Goodrich JA, Wang H, Jia Q, et al.
Environment International, 2024.

The Potential Mediating Role of the Gut Microbiome and Metabolites in the Association Between PFAS and Kidney Function in Young Adults: A Proof-of-Concept Study.

Hampson HE, Li S, Walker DI, Wang H, Jia Q, et al.
Science of The Total Environment, 2024.

PRESENTATIONS

Praxis-BGM: A Prior-Augmented and Regularized Natural-Gradient Variational Inference on Gaussian Mixture Models

Joint Statistical Meetings (JSM), Nashville, TN, Aug. 2025

BiTr-UNet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation

MICCAI BrainLes Workshop, Virtual, Sep. 2021